North Dakota K-12 AI Guidance Framework

Artificial Intelligence (AI) is the latest in a long and continuing list of tools that can be used to transform teaching, streamline business operations, and personalize learning. Implementing AI (as with any new tool) requires thoughtful planning and alignment with existing educational goals, values, and priorities. We must keep the main thing the main thing: the preparation of our young learners for their next challenges and goals.

As your institution begins to implement AI tools, please remember that AI cannot exist on its own. It requires human input (in the form of data, questions or prompts, and preprogrammed algorithms or mathematical instructions). This technology can process large volumes of data with incredible speed, efficiency, and precision to produce an output. Humans should then evaluate this output to look for correctness, applicability, rationality, and bias. This is the Human – Technology – Human workflow that is essential to the ethical function of AI.

AI is best used to augment uniquely human traits, such as creativity, while automating more mundane tasks.

Together, we will create an educational environment where technology is supported, where human control and inquiry lead to limitless learning, and where our students are ready to lead us into a world boosted by AI.

General Resources

Who	What	Link
Code.org	Lesson plans and guidance	Learn today, build a brighter tomorrow. | Code.org
Creative commons	Creative Commons is an international nonprofit organization that empowers people to grow and sustain the thriving commons of shared knowledge and culture we need to address the world's most pressing challenges and create a brighter future for all.	Homepage - Creative Commons
Cyber.org	Lesson plans and guidance	Empowering Educators to Teach Cyber | Cyber.org
ISTE (International Society for Technology in Education)	Free guides for engaging students in AI creation	ISTE | Artificial Intelligence in Education
TeachAI	Policy and Guidance	TeachAI

Teacher Resources

Who	What
Brisk Teaching	Content generation
Diffit	Differentiated resources for various reading levels
Curipod	AI based question creation and more
Almanack AI	Lesson Plans and classroom resources
EduAide AI	Content generation
Magic School	Content generation

Definitions

Term	Definition	Source
Accessibility	Accessibility is when the needs of people with disabilities are specifically considered, and products, services, and facilities are built or modified so that they can be used by people of all abilities	Disability and Health Inclusion Strategies | CDC
Adaptive Learning	Adaptive learning allows the course material to be customized to the learner, which creates a unique	Adaptive Learning: What is It, What are its Benefits and How

	experience not available in traditional classes. Technology-based adaptive learning systems or e-learning systems can provide students with immediate assistance, resources specific to their learning needs, and relevant feedback that students may need.	Does it Work? - Educational Technology
Algorithm	Algorithms are the “brains” of an AI system and what determines decisions in other words, algorithms are the rules for what actions the AI system takes. Machine learning algorithms can discover their own rules (see Machine learning for more) or be rule-based where human programmers give the rules	Glossary of Artificial Intelligence Terms for Educators – CIRCLS
Artificial Intelligence	AI is a branch of computer science. AI systems use hardware, algorithms, and data to create “intelligence” to do things like make decisions, discover patterns, and perform	Glossary of Artificial Intelligence Terms for Educators – CIRCLS

	some sort of action. AI is a general term and there are more specific terms used in the field of AI. AI systems can be built in different ways, two of the primary ways are: (1) through the use of rules provided by a human (rule-based systems); or (2) with machine learning algorithms. Many newer AI systems use machine learning	
Asynchronous Learning	Asynchronous learning is any type of learning that you undertake on your own schedule and which does not require consistent real-time interactions with an instructor. It differs from synchronous learning, which can be done online or in-person, and typically requires you and your classmates to attend scheduled classes with your instructor.	What Is Asynchronous Learning? | Coursera
Blended Learning	The term blended learning is generally applied to the practice of using both online and in-	Blended Learning Definition (edglossary.org)

	person learning experiences when teaching students. In a blended-learning course, for example, students might attend a class taught by a teacher in a traditional classroom setting, while also independently completing online components of the course outside of the classroom	
Copyright	Copyright is a type of intellectual property that protects original works of authorship as soon as an author fixes the work in a tangible form of expression.	What is Copyright? | U.S. Copyright Office
Creative Commons	Creative Commons (CC) is a set of licenses that creators can use to allow the public to use their work, subject only to the restrictions they choose. For instance, the creator could choose a license that allows their work to be re-used for commercial purposes (advertising and selling	Creative Commons Guide | Digital Citizenship+ Resource Platform (harvard.edu)

	things), or one that doesn't	
Digital Citizenship	Digital citizenship is the ability to navigate our digital environments in a way that's safe and responsible and to actively and respectfully engage in these spaces.	What is Digital Citizenship? | MediaSmarts
Digital Learning	Digital learning refers to any sort of learning that makes use of and takes advantage of technology	Digital Learning | (edapp.com)
Digital Literacy	Digital literacy means having the skills to effectively use technology, and the knowledge and skills to do so safely and responsibly. "Digital" refers to technology, ranging from computers and the internet to technological objects and programs such as cellphones, smart home systems, check-in kiosks at airports and more.	What is Digital Literacy: Definition and Uses in Daily Life | Learning.com
Large Language Model	Large language models (LLMs) Large language models form the foundation for generative AI (GenAI) systems.	Glossary of Artificial Intelligence Terms for

	GenAI systems include some chatbots and tools including OpenAI's GPTs, Meta's LLaMA, xAI's Grok, and Google's PaLM and Gemini. LLMs are artificial neural networks. At a very basic level, the LLM detected statistical relationships between how likely a word is to appear following the previous word in their training. As they answer questions or write text, LLM's use the model of the likelihood of a word occurring to predict the next word to generate. LLMs are a type of foundation model, which are pre-trained with deep learning techniques on massive data sets of text documents. Sometimes, companies include data sets of text without the creator's consent.	Educators – CIRCLS
Machine Learning	Machine learning is a field of study with a range of approaches to developing algorithms that can be used in AI systems. AI is	Glossary of Artificial Intelligence Terms for

	a more general term. In ML, an algorithm will identify rules and patterns in the data without a human specifying those rules and patterns. These algorithms build a model for decision making as they go through data. (You will sometimes hear the term machine learning model.) Because they discover their own rules in the data they are given, ML systems can perpetuate biases. Algorithms used in machine learning require massive amounts of data to be trained to make decisions.	Educators – CIRCLS

Implementation Roadmap

Inspired by the North Carolina guidance and North Dakota's EduTech SCRIPT training

This generic implementation roadmap can be used when implementing any large technology project

Establish and Evaluate Foundation

- Host a meeting with key stakeholders (school leaders and educators, administrators, student leaders, school board

members)
- Evaluate the district's current educational goals, values, and priorities and update as needed
- Create a team to develop/update AI academic policy and guidelines (or update current guidelines to include AI)

Evaluate Technology Infrastructure

- Review current education technology (EdTech) to ensure it can support updated policies and guidelines
- Review current EdTech providers supplying AI to vet their safety, privacy, reliability, and efficacy to determine if they are appropriate for your school and which users they will support based on their terms of service

Develop Staff

- Provide targeted professional development for educators on AI, including its impact, effective use, capabilities, limitations, concerns and responsible generative AI use
- Share AI guidelines draft for feedback; work with teachers on what the guidelines mean for their classrooms
- Support teachers in updating their syllabi and/or classroom policies to include AI integrity guidelines that align with district/school guidelines
- Work with teachers to help them rethink plagiarism and academic integrity in the AI age and support them in shifting assessments to AI-resistant, AI-assisted, and AI-partnered versions

Educate Students and Community

- Share AI guidelines at schoolwide events, including with parents and guardians, to build a common understanding

- Teachers review guidelines in each classroom along with syllabi and examples of appropriate and inappropriate student use
- Implement AI training to upskill students and ensure they are prepared to mitigate any biases, inaccuracies, or issues that may arise and utilize generative AI effectively as a learning partner
- Provide content reviews and ongoing opportunities for training and learning to teachers and the school community

Assess and Progress

- Create a plan for constant review and reevaluation of academic guidelines in light of AI evolution and advances
- Evaluate new AI tools for appropriateness to launch pilot programs – rubrics to evaluate tools
- Continuously update and train across school community, including sharing exemplars and providing opportunities to express concerns
- Elevate best practices for generative AI implementation from across community and partners

Implementation Checklist

School Leadership	
Educators	
Business and IT Operations	
Policy, Ethical and Legal Considerations	

Public Outreach

What is AI? A history so far

Artificial Intelligence emerged as a field of computer science in the 1950s. Prior to that, computers lacked a key prerequisite for intelligence: they couldn't store commands, they could only execute them. Computers were also extremely expensive; the cost of leasing a computer ran up to $200,000 a month.

In 1955, Allen Newel, Cliff Shaw, and Herbert Simon developed the Logic Theorist, a program solving program designed to mimic the skills of a human. In 1956, John McCarthy and Marvin Minsky hosted the 'Dartmouth Summer research Project on Artificial Intelligence' (the term Artificial Intelligence was coined at that event).-

From 1957 to 1974, AI flourished. Computers could store more information and became faster, cheaper, and more accessible. Machine Learning (ML) emerged as a technique where computers were trained to improve their performance by processing large amounts of data. These ML algorithms improved, and people got better at knowing which algorithm to apply to their problem. Early educational applications included intelligent tutoring systems and adaptive learning.

In the 1980's, AI was reignited by two sources: an expansion of the algorithmic toolkit, and a boost of funds. John Hopfield and David Rumelhart popularized Deep Learning (DL) techniques which allowed computers to learn using experience. Deep Learning is a Machine Learning technique that utilized neural networks and algorithms inspired by how human brains learn and process information.

During the 1990s and 2000s, many of the landmark goals of artificial intelligence had been achieved. One of the issues holding up progress, limits to computer storage, was no longer a problem. In 1997, reigning world chess champion and grand master Gary Kasparov was defeated by IBM's *Deep Blue*, a chess playing

computer program. In the same year, speech recognition software, developed by Dragon Systems, was implemented on *Windows*. This was another great step forward but in the direction of the spoken language interpretation endeavor.

We now live in the age of "big data" where we have the capacity to collect huge sums of information too cumbersome for a person to process. The application of artificial intelligence in this regard has already been quite fruitful in several industries such as technology, banking, marketing, and entertainment. The Large Language Models (LLMs) is a product of DL techniques, are models specialized for tasks like natural language processing, text generation and translation.

Generative AI (Gen AI) is a powerful category of AI that includes LLMs and other models that generate text, images, videos, or music. The internal workings of Gen AI models can lack transparency and explainability, making it challenging to build trust and ensure accountability. Additional issues specific to Gen AI in education include bias, misinformation, and overreliance on AI tools.

We can't be sure what the future holds, but we can be fairly certain these technologies will continue to grow. We must educate our students to become AI literate. AI literacy encompasses understanding how AI works, using AI responsibly, and recognizing its social and ethical impacts. It includes understanding AI's potential benefits and risks and how to mitigate the risks. AI literacy is a crucial aspect of digital competence and equips individuals to engage productively and responsibly with AI in society, the economy, and their personal lives. Championing AI literacy is essential to prepare students to be informed citizens who can also thrive in a future workforce in which AI is ubiquitous across all fields.

Anyoha, Rockwell. "The History of Artificial Intelligence", 28 Aug. 2017, The History of Artificial Intelligence - Science in the News (harvard.edu)

AI: Strengths and Weaknesses

These examples illustrate the strengths and limitations of AI in various domains. As AI technology continues to advance, some of these limitations may be addressed, while new applications for AI's strengths will likely emerge.

AI is good at:

- Data Analysis:
  - Financial forecasting. AI analyzes market trends to predict stock performance.
  - Course scheduling.
  - Bus route planning.
- Pattern Recognition:
  - Fraud detection. AI monitors transactions to detect unusual patterns indicative of fraud.
  - Weather forecasting. AI uses atmospheric data to recognize patterns and predict weather changes.
- Predictive Analytics:
  - Customer behavior. AI predicts future purchasing patterns based on past behavior.
  - Equipment maintenance. AI anticipates when industrial equipment will need maintenance.
- Natural Language Processing (NLP):
  - Translation services. AI translates languages in real time.
  - Sentiment analysis. AI assesses public sentiment from social media posts.
- Image and Speech Recognition:
  - Facial recognition. AI identifies individuals in security footage.
  - Voice assistants: AI, such as Siri and Alexa, recognizes and responds to voice commands.

- Automating Repetitive Tasks:
  - Manufacturing robots. AI controls robots that perform assembly line tasks.
  - Data entry. AI automates the input of data into systems.
- Playing Games:
  - Chess programs: AI, such as Deep Blue or AlphaZero, can defeat human grandmasters.
  - Video games. AI non-player characters (NPCs) adapt to player actions.

AI is Not Good At:

- Understanding Context:
  - Sarcasm detection. AI often misinterprets sarcasm as literal speech.
  - Complex decision making. AI struggles with decisions that require understanding of broader implications.
- Creative Thinking:
  - Inventing new concepts. AI cannot invent genuinely new concepts or ideas.
  - Art creation. While AI can generate art, it lacks the personal touch and originality of a human artist.
- Emotional Intelligence:
  - Empathy. AI cannot truly empathize with human emotions.
  - Therapy and counseling. AI lacks the nuanced understanding required for effective mental health support.
- General Problem Solving:
  - Interdisciplinary research. AI finds it difficult to integrate knowledge across different fields.
  - Unstructured tasks. AI struggles with tasks that lack clear rules or objectives.
- Ethical Decision Making:
  - Legal judgments. AI cannot make legal decisions that require ethical considerations.

    - Moral dilemmas. AI is not equipped to handle complex moral dilemmas.
- Physical Tasks Requiring Fine Motor Skills:
    - Surgery. While there are robotic aids, AI alone cannot perform delicate surgeries.
    - Craftsmanship. AI cannot match the dexterity and skill of a human craftsperson.

Student Grade Levels

There is no avoiding AI; it has quickly become integrated into so many things we interact with daily. Because of this, there is a growing consensus that students need to be AI-literate by the time they enter the workforce. This creates issues for educators who need to balance the introduction of these technologies with the understanding of student's cognitive development.
Below are some specific things to think about as you introduce these new technologies to varying grade levels within your district.

Lower Elementary (K-2): These students need to understand that AI is not a real person. Kindergartners through 2nd graders are at a point in their brain development where they are more likely to attribute human qualities to artificially intelligent technologies like smart speakers and chatbots. They may even trust what an AI-powered device or tool is saying over the adults in their lives, like teachers. Be careful about using language that humanizes AI tools.

Upper Elementary (3-5): Trying and failing (or not succeeding) is another crucial part of learning. Technology, powered by AI, can be good at answering questions, but an overreliance on that kind of technology can short-circuit students' development of problem-solving

skills. As you introduce these technologies, don't lose focus on developing problem-solving skills.

Middle School (6-8): You should build on students' developing critical and abstract thinking skills by having them critique AI outputs. Exercises in which students ask a generative AI chatbot to answer a question or write an essay and then critique it—looking for factual errors and the like—would be developmentally appropriate for this age group. You should also note any age-level restrictions on some AI products. Some tools are prohibited for children under age 13 or require parental/guardian permission.

High School (9-12): Rather than police AI use, teach students about how AI works and its limitations, including the effects of bias, stereotypes, and inaccuracies. Teach them a healthy skepticism. Teachers should be judicious about how many bright, new shiny AI tools they bring to the classroom, especially as more and more are developed every day.

Prothero, Arianna, "*What is Age-Appropriate Use of AI? 4 Developmental Stages to Know About*" 19 Feb 2024. https://www.edweek.org/technology/what-is-age-appropriate-use-of-ai-4-d...

AI for Accessibility

Some students often require individualized educational programs to meet their specific needs. The application of AI can help meet some of these specific needs. AI-powered tools and technologies can provide personalized learning experiences, aid in skill development, and improve accessibility for students with special needs.

The concept of accessibility is also important for the individual needs of learners, such as English language learners, learners in rural communities, and learners from economically disadvantaged students.

Personalized Learning: AI can help in designing **personalized learning** that adapts to the learning pace and style of each student. Through advanced algorithms and machine learning, AI can analyze each student's learning patterns, strengths, and weaknesses, and adapt the educational content accordingly.

Assistive Technology: AI can augment assistive technologies. For instance, speech recognition can help students with speech impairments communicate effectively. AI tools can also help transcribe speech into text in real time helping student with hearing impairments follow along in class. Predictive text tools can also assist students with dyslexia in writing.

Data Collection and Analysis: AI can also assist in data collection and analysis, helping educators understand student performance and devise effective teaching strategies. AI-powered predictive analytics can help educators to identify potential learning obstacles and provide timely intervention. By analyzing data on student performance and behavior, AI can predict which students might be at risk of falling behind and suggest strategies to help them catch up. This proactive approach can lead to better learning outcomes and a more inclusive educational environment.

By providing personalized learning experiences, developing assistive technologies, and improving accessibility, AI is poised to revolutionize special education. As we continue to explore the potential of AI in this

field, it's important for educators, parents, and students to stay informed and embrace the opportunities that this technology brings.

Admin. "*Exploring the Role of AI in Special Education*". 29 Feb 2024. https://www.21kschool.com/us/blog/ai-in-special-education/

This guidance could not have been developed without the expertise and knowledge of our writing team, made up of educators, students, and stakeholders.